

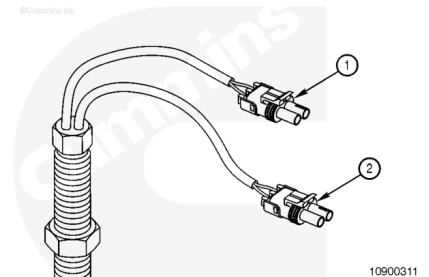
Trainings ▾

Initial Check

There are two connections coming from the sensor, one is for the engine tachometer and the other is for the air intake heater system.

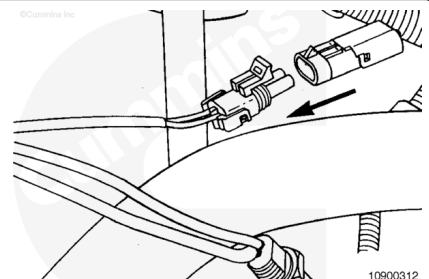
1. White and green: to engine wiring harness
2. Red and black: to air heater control harness.

Check to verify they are connected.

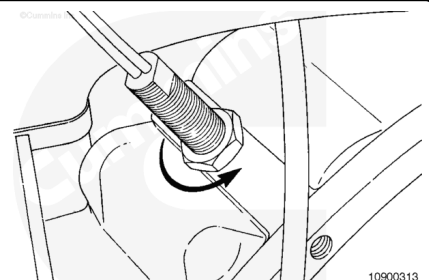


Remove

Disconnect the engine speed sensor from the engine wiring harness.



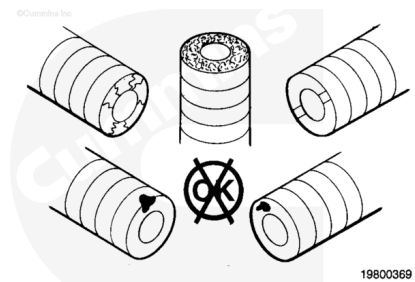
Remove the sensor from the flywheel housing.



Inspect for Reuse

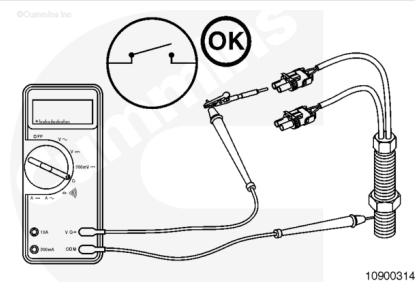
Inspect the engine speed sensor for debris, cracked or chipped potting, and damage from contact with the flywheel.

Clean or replace the engine speed sensor if necessary.

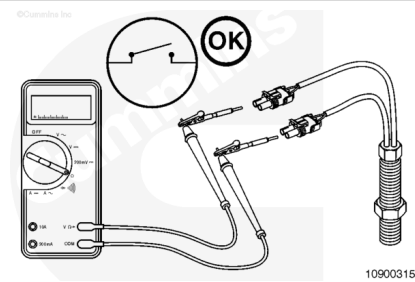


Test

1. Short Circuit to Ground: Measure the resistance of each connector (Pin A or B) of both circuits to ground (sensor body) with a multimeter. The multimeter **must** show an open circuit (100k ohms or more). If fewer than 100k ohms are measured, there is a short circuit to the sensor. Replace the sensor.

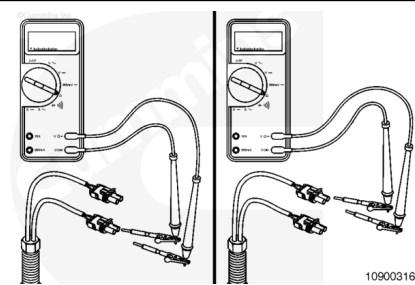


2. Short Circuit Between Coils: Check the resistance between connectors (Pin A to Pin A and Pin B to Pin B) with a multimeter. The multimeter **must** show an open circuit (100k ohms or more). If fewer than 100k ohms are measured, there is a short circuit in the sensor. Replace the sensor.



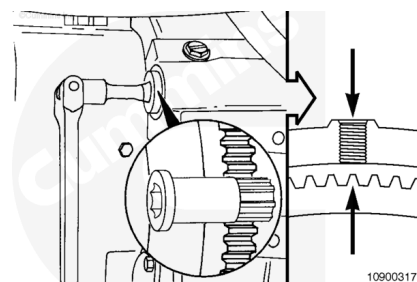
3. Coil Resistance: Check the resistance between the pins of each connector (Pin A to Pin B).

Resistance Specification	
First coil	750 to 1100 ohms
Second coil	1100 to 1500 ohms



Install

Use a barring gear, Part Number 3377371, to rotate the crankshaft so that a ring gear tooth is at the center of the hole for the engine speed sensor.

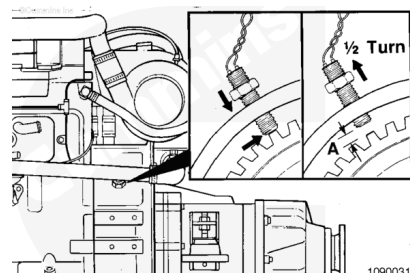


Insert the engine speed sensor and turn it until the end touches the ring gear tooth.

Back off a half turn and tighten the locknut while holding the engine position sensor.

Torque Value: 34 to 47 n·m [25 to 35 ft·lb]

Overtightening will damage the sensor.



Install the engine speed sensor receptacle with red and black leads on the air intake heater control harness plug.

The white and green wire plug is connected to the wiring harness for the tachometer sensing.

1. White and green: to engine wiring harness
2. Red and black: to air heater control harness.

